WORKED SOLUTION

STEP II 1987, Question 1

Pure • Algebraic Identities

STEP II 1987, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/87-s2-q1

The key idea

Don't try to substitute for one of the letters. Expand the right-hand side instead. The squares a^2, b^2, c^2 appear on both sides and cancel, so what is left is x times a linear expression, and that expression is exactly a multiple of the given condition. Once you see where the numbers 7 and 11 come from, the general result follows.

Solution

It helps to do the expansion once, in general. For any numbers p, q, r,

$$(px-a)^2+(qx-b)^2+(rx-c)^2=(p^2+q^2+r^2)x^2-2x(pa+qb+rc)+a^2+b^2+c^2,$$

so

$$(px-a)^2+(qx-b)^2+(rx-c)^2-(a^2+b^2+c^2)=x[(p^2+q^2+r^2)x-2(pa+qb+rc)]. \qquad (*)$$

Part (i)

Here $(p,q,r)=(1,2,3)$, so $p^2+q^2+r^2=14$ and $(*)$ becomes

$$(x-a)^2+(2x-b)^2+(3x-c)^2-(a^2+b^2+c^2)=x[14x-2(a+2b+3c)].$$

If $a+2b+3c=7x$, the bracket is $14x-14x=0$. So the two sides are equal, as required.

Part (ii)

Now $(p,q,r)=(2,3,3)$, so $p^2+q^2+r^2=4+9+9=22$ and

$$(2x-a)^2+(3x-b)^2+(3x-c)^2-(a^2+b^2+c^2)=x[22x-2(2a+3b+3c)].$$

If $2a+3b+3c=11x$, the bracket is $22x-22x=0$, which proves (ii).

Ask yourself Why 7 in (i) and 11 in (ii)?

They are half of $p^2+q^2+r^2$: $7=\frac{1}{2}(1+4+9)$ and $11=\frac{1}{2}(4+9+9)$. That tells us what the general statement must be.

The general result

Identity $(*)$ proves the following directly.

For any real numbers p,q,r,a,b,c,x, if

$$2(pa+qb+rc)=(p^2+q^2+r^2)\,x,$$

then

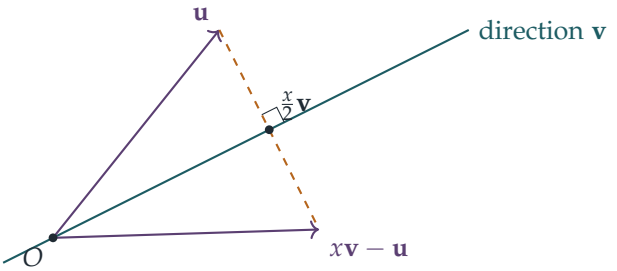
$$a^2+b^2+c^2=(px-a)^2+(qx-b)^2+(rx-c)^2.$$

Part (i) is the case $(p,q,r)=(1,2,3)$ and part (ii) is the case $(p,q,r)=(2,3,3)$.

The same expansion works with any number of variables. If $2\sum_{i=1}^{n} p_i a_i=(\sum_{i=1}^{n} p_i^2)x$, then $\sum a_i^2=\sum(p_i x-a_i)^2$. From $(*)$ we also get a converse: if $x\neq 0$ and the identity holds, then the condition must hold.

What is going on geometrically

Write $\mathbf{u}=(a,b,c)$ and $\mathbf{v}=(p,q,r)$. The condition reads $\mathbf{u}\cdot\mathbf{v}=\frac{x}{2}|\mathbf{v}|^2$, which says that $\frac{x}{2}\mathbf{v}$ is the foot of the perpendicular from $\mathbf{u}$ to the line through O in direction $\mathbf{v}$. The vector $x\mathbf{v}-\mathbf{u}=2\cdot\frac{x}{2}\mathbf{v}-\mathbf{u}$ is then the reflection of $\mathbf{u}$ in that line. A reflection preserves length, so $|x\mathbf{v}-\mathbf{u}|=|\mathbf{u}|$, and that is the identity.



Checks

Try (i) with $x=1$, $a=b=0$, $c=\frac{7}{3}$. The left side is $\frac{49}{9}$ and the right side is $1+4+(\frac{2}{3})^2=5+\frac{4}{9}=\frac{49}{9}$. We also tested both parts and the general result in exact rational arithmetic, on 2000 random cases each (up to six variables in the general case), and confirmed that the identity fails whenever $x\neq 0$ and the condition fails.

Common slips

- Substituting $c = \frac{1}{3}(7x - a - 2b)$ straight away. It works, but the algebra is messy and hides the structure.
- Giving the general result as "$pa + qb + rc = kx$" with no link between k and p, q, r. The identity needs $k = \frac{1}{2}(p^2 + q^2 + r^2)$.

Beyond STEP

The geometric picture is the start of linear algebra. The map $\mathbf{u} \mapsto 2\frac{\mathbf{u}\cdot\mathbf{v}}{|\mathbf{v}|^2}\mathbf{v} - \mathbf{u}$ is a reflection, an orthogonal linear map, and orthogonal maps are exactly the linear maps that preserve lengths. Projections and reflections like this one underlie least squares and the Householder transformations used in numerical computation. Strang's *Introduction to Linear Algebra* develops these ideas with the same geometric flavour.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.